

ECE 3441 Spring 2025

Quiz 2

Version A

- Closed book, closed notes
- No cheat sheet is allowed.
- Only four function calculators are allowed.
- Show all details of your work to obtain credit (The answer alone will not suffice).
- You may not talk during the quiz to others or to yourself.
- You have 60 minutes.

Please do not open booklet unless instructed to do so, or substantial penalty will result.

SOLUTIONS

First Name Last Name, Peoplesoft ID, Date (YYYY/MM/DD)

Question	Points	Maximum
1		30
2		20
3		15
4		25
5		20

Total: _____/(100 possible points + 10 bonus points)

Question #1:

Given the following equation:

$$F(A, B, C, D) = \sum m(6, 13, 14, 15) + \sum d(2, 5, 7, 9, 11)$$

Find the minimum POS by only using a K-Map:

$$F = \sum m(0110, 1101, 1110, 1111) + \sum d(0010, 0101, 0111, 1001, 1011)$$

AB \ CD	00	01	11	10
00	0	0	0	X
01	0	X	X	1
11	0	1	1	1
10	0	X	X	0

$$F = (C + D)(B)$$

Question #2:

Show whether the following is true or false. Hint: truth table; K-Maps might be harder.

$$A'B'C' + A'B'C + A'BC' + AB'C = (A + B' + C')(A' + B + C)(A' + B' + C)(A' + B' + C')$$

$$\underbrace{000 \ 001 \ 010 \ 101}_{F_1} \quad \underbrace{011 \ 100 \ 110 \ 111}_{F_2}$$

A	B	C	F ₁	F ₂
0	0	0	1	
0	0	1	1	
0	1	0	1	
0	1	1		0
1	0	0		0
1	0	1	1	
1	1	0		0
1	1	1		0

$$\therefore F_1 = F_2$$

TRUE

Question #3:

Each of three coins has two sides, heads and tails. Represent the heads or tails status of each coin by a logical variable (A for the first coin, B for the second coin, and C for the third) where the logical variable is 1 for heads and 0 for tails. Write a logic function $F(A, B, C)$ which is 1 only if exactly one of the coins is heads after a toss of the coins.

Express F :

a) As a minterm expansion:

$$\begin{aligned} F &= AB'C' + A'BC' + A'B'C \\ &= 100 + 010 + 001 \\ &= \sum m(1, 2, 4) \end{aligned}$$

b) As a maxterm expansion:

$$\begin{aligned} F &= \prod M(0, 3, 5, 6, 7) \\ &= (A+B+C)(A+B'+C')(A'+B+C')(A'+B'+C) \\ &\quad \dots (A'+B'+C') \end{aligned}$$

Question #4:

Using K-Maps, given $F(A, B, C) = A(B + C')$:

- a) Express F as a minterm expansion (use m -notation and $\sum m(\dots)$ notation)

$$F = AB + AC'$$

AB \ C	0	1
00	0	0
01	0	0
11	1	1
10	1	0

$$F = 100 + 110 + 111$$

$$= \sum m(4, 6, 7)$$

- b) Express F as a maxterm expression (use M -notation and $\prod M(\dots)$ notation)

$$F = \prod M(0, 1, 2, 3, 5)$$

Question #5:

Find the minimum SOP of the following expression using only K-Maps.

$$F(A, B, C, D) = B'C'D' + BD + CD' + B'CD$$

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11	0	1	1	1
10	1	0	1	1

$$F = B'D' + BD + C$$